WORKED SOLUTION

STEP III 2016, Question 3

Pure • Integration and Polynomials

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s3-q3

This question has a full guided treatment in *Beyond STEP*, Section 4.6, *Integrals With No Formula*: https://beyondstep.keithmcnulty.org/ch04/no-formula.html

The key idea

An equation of the form $\int g(x)\,\mathrm{e}^x\,\mathrm{d}x = \frac{P}{Q}\,\mathrm{e}^x + \text{constant}$ is just a statement about a derivative, so differentiate the right-hand side and clear denominators. That turns the question into a polynomial identity, and polynomial identities can be attacked in two ways: by substituting a well-chosen value (here $x = -1$) and by comparing degrees.

Solution

Throughout, write P, Q for $P(x)$, $Q(x)$. By the quotient and product rules,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{P}{Q}\,\mathrm{e}^x\right) = \frac{PQ + P'Q - PQ'}{Q^2}\,\mathrm{e}^x.$$

So the integral equation holds exactly when this equals the integrand (with Q not the zero polynomial).

Part (i)

Equating the derivative with $\frac{x^3-2}{(x+1)^2}\,\mathrm{e}^x$, cancelling $\mathrm{e}^x \neq 0$ and multiplying up gives the polynomial identity

$$(x+1)^2\left(PQ + P'Q - PQ'\right) = (x^3 - 2)\,Q^2. \qquad (*)$$

Q has a factor $x+1$. Put $x = -1$ in $(*)$. The left-hand side is 0, and the right-hand side is $-3\,Q(-1)^2$. Hence $Q(-1) = 0$, and by the factor theorem $x+1$ divides Q.

Degrees. P is not the zero polynomial, since otherwise the derivative would be 0, not the integrand. Let $\deg P = p$ and $\deg Q = q$, with leading coefficients $a \neq 0$ and

$b \neq 0$. In $PQ + P'Q - PQ'$, the term PQ has degree $p+q$ with leading coefficient ab, while $P'Q$ and PQ' each have degree at most $p+q-1$. So $PQ + P'Q - PQ'$ has degree exactly $p+q$, and the left-hand side of $(*)$ has degree $p+q+2$. The right-hand side has degree $3+2q$. Therefore

$$p+q+2 = 2q+3 \quad \Longrightarrow \quad p = q+1,$$

so the degree of P is exactly one more than that of Q.

The case $Q = x+1$. Now P has degree 2, say $P = ax^2 + bx + c$. With $Q = x+1$ and $Q' = 1$, $(*)$ becomes

$$(x+1)^2\big((x+1)P + (x+1)P' - P\big) = (x^3-2)(x+1)^2,$$

and cancelling $(x+1)^2$ gives

$$xP + (x+1)P' = x^3 - 2.$$

Since $xP = ax^3 + bx^2 + cx$ and $(x+1)P' = (x+1)(2ax+b) = 2ax^2 + (2a+b)x + b$,

$$ax^3 + (2a+b)x^2 + (2a+b+c)x + b = x^3 - 2.$$

Comparing coefficients: $a = 1$; the constant term gives $b = -2$; then the x^2 term, $2a+b = 0$, is consistent; and the x term gives $c = -(2a+b) = 0$. Hence

$$P(x) = x^2 - 2x, \qquad \int \frac{x^3-2}{(x+1)^2}\,\mathrm{e}^x\,\mathrm{d}x = \frac{x^2-2x}{x+1}\,\mathrm{e}^x + \text{constant}.$$

Part (ii)

Suppose, for a contradiction, that such P and Q exist with no common factor. As before, differentiating and clearing denominators gives

$$(x+1)\big(PQ + P'Q - PQ'\big) = Q^2. \qquad (**)$$

Putting $x = -1$ gives $Q(-1)^2 = 0$, so $x+1$ divides Q. Write

$$Q = (x+1)^n R, \qquad n \geq 1, \quad R(-1) \neq 0,$$

so that n is the exact power of $x+1$ in Q. Then $Q' = n(x+1)^{n-1}R + (x+1)^n R'$, and

$$\begin{aligned} PQ + P'Q - PQ' &= (x+1)^n\big(PR + P'R - PR'\big) - n(x+1)^{n-1}PR \\ &= (x+1)^{n-1}\Big[(x+1)\big(PR + P'R - PR'\big) - nPR\Big]. \end{aligned}$$

Substituting into $(**)$ and cancelling the common factor $(x+1)^n$ from both sides (the right-hand side is $(x+1)^{2n}R^2$):

$$(x+1)\big(PR + P'R - PR'\big) - nPR = (x+1)^n R^2.$$

Ask yourself Why go to the trouble of pulling out the exact power of $x+1$, rather than just substituting $x=-1$ into $(**)$ again?

Substituting $x=-1$ straight into $(**)$ only tells us $Q(-1)=0$; every term then vanishes and we learn nothing about P. Removing all the factors of $x+1$ from Q leaves a term, $-nPR$, that does *not* automatically vanish at $x=-1$.

Now put $x=-1$. Since $n\geq 1$, the right-hand side is 0, and so is the first term on the left. Hence

$$-n\,P(-1)\,R(-1)=0.$$

But $n\neq 0$ and $R(-1)\neq 0$, so $P(-1)=0$. Then $x+1$ is a factor of P as well as of Q, contradicting the assumption that they have no common factor. So no such polynomials exist. ■

(The restriction to coprime P, Q costs nothing: any common factor could be cancelled from P/Q without changing the function.)

Checks

- Differentiate the answer to (i) directly: $\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{x^2-2x}{x+1}\right)=\frac{x^2+2x-2}{(x+1)^2}$, and adding $\frac{x^2-2x}{x+1}=\frac{x^3-x^2-2x}{(x+1)^2}$ gives $\frac{x^3-2}{(x+1)^2}$, as required.
- Numerically, $\int_0^1 \frac{x^3-2}{(x+1)^2}\,\mathrm{e}^x\,\mathrm{d}x\approx -1.359141$, which matches $\left[\frac{x^2-2x}{x+1}\mathrm{e}^x\right]_0^1=-\frac{\mathrm{e}}{2}$.
- In (i) the degrees agree: P has degree 2, one more than $Q=x+1$.

Common slips

- Forgetting that $P'Q-PQ'$ has degree *less* than $p+q$, because the leading terms cancel; it is PQ alone that fixes the degree.
- In (ii), substituting $x=-1$ only into $(**)$ and stopping once $Q(-1)=0$ is found. The contradiction needs the exact power of $x+1$ in Q.

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Part (ii) says that $\int \frac{\mathrm{e}^x}{x+1}\,\mathrm{d}x$, essentially the exponential integral Ei, is not "elementary" in this restricted sense. The full theory, which decides when an integral can be written using elementary functions at all, is Liouville's theorem and the Risch algorithm of differential algebra; the argument here is a small instance of it. For a careful treatment of the polynomial algebra (factor theorem, divisibility, coprimality) used above, see Allenby, *Rings, Fields and Groups*.

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